

Ex.No:4

**EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION,
FILTERING, STATISTICAL ANALYSIS AND CORRELATION**

Date:

AIM:

To perform data inspection, filtering, grouping, aggregation, descriptive statistical analysis, and correlation analysis using Pandas and visualize the results using Matplotlib.

ALGORITHM:

CODE:

```
import pandas as pd  
df=pd.read_csv("C:/Users/DELL/Downloads/Sleep_health_and_lifestyle_dataset.csv")  
print(df.head())
```

	Person ID	Gender	Age	Occupation	Sleep Duration	\
0	1	Male	27	Software Engineer	6.1	
1	2	Male	28	Doctor	6.2	
2	3	Male	28	Doctor	6.2	
3	4	Male	28	Sales Representative	5.9	
4	5	Male	28	Sales Representative	5.9	

	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	\
0	6	42	6	Overweight	
1	6	60	8	Normal	
2	6	60	8	Normal	
3	4	30	8	Obese	
4	4	30	8	Obese	

	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	126/83	77	4200	NaN
1	125/80	75	10000	NaN
2	125/80	75	10000	NaN
3	140/90	85	3000	Sleep Apnea
4	140/90	85	3000	Sleep Apnea

```
print(df.shape)
```

```
(374, 13)
```

```
print(df.dtypes)
```

```

Person ID          int64
Gender            object
Age              int64
Occupation        object
Sleep Duration    float64
Quality of Sleep  int64
Physical Activity Level int64
Stress Level      int64
BMI Category      object
Blood Pressure    object
Heart Rate        int64
Daily Steps       int64
Sleep Disorder    object
dtype: object

```

```

high_sleep = df[df["Sleep Duration"] > 7]
print(high_sleep)

```

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep \
7	8	Male	29	Doctor	7.8	7
8	9	Male	29	Doctor	7.8	7
9	10	Male	29	Doctor	7.8	7
11	12	Male	29	Doctor	7.8	7
19	20	Male	30	Doctor	7.6	7
..	...	...	...	...	...	...
369	370	Female	59	Nurse	8.1	9
370	371	Female	59	Nurse	8.0	9
371	372	Female	59	Nurse	8.1	9
372	373	Female	59	Nurse	8.1	9
373	374	Female	59	Nurse	8.1	9

	Physical Activity Level	Stress Level	BMI Category	Blood Pressure \
7	75	6	Normal	120/80
8	75	6	Normal	120/80
9	75	6	Normal	120/80
11	75	6	Normal	120/80
19	75	6	Normal	120/80
..	...	...	...	...
369	75	3	Overweight	140/95
370	75	3	Overweight	140/95
371	75	3	Overweight	140/95
372	75	3	Overweight	140/95
373	75	3	Overweight	140/95

	Heart Rate	Daily Steps	Sleep Disorder
7	70	8000	NaN
8	70	8000	NaN
9	70	8000	NaN
11	70	8000	NaN
19	70	8000	NaN
..	...	...	...
369	68	7000	Sleep Apnea
370	68	7000	Sleep Apnea
371	68	7000	Sleep Apnea
372	68	7000	Sleep Apnea
373	68	7000	Sleep Apnea

[219 rows x 13 columns]

```
female = df[df["Gender"] == "Female"]
print(female)
```

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	\
16	17	Female	29	Nurse	6.5	5	
18	19	Female	29	Nurse	6.5	5	
30	31	Female	30	Nurse	6.4	5	
31	32	Female	30	Nurse	6.4	5	
32	33	Female	31	Nurse	7.9	8	
..	...	...	...	...	...	...	
369	370	Female	59	Nurse	8.1	9	
370	371	Female	59	Nurse	8.0	9	
371	372	Female	59	Nurse	8.1	9	
372	373	Female	59	Nurse	8.1	9	
373	374	Female	59	Nurse	8.1	9	

	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	\
16	40	7	Normal Weight	132/87	
18	40	7	Normal Weight	132/87	
30	35	7	Normal Weight	130/86	
31	35	7	Normal Weight	130/86	
32	75	4	Normal Weight	117/76	
..	...	...	...	...	
369	75	3	Overweight	140/95	
370	75	3	Overweight	140/95	
371	75	3	Overweight	140/95	
372	75	3	Overweight	140/95	
373	75	3	Overweight	140/95	

	Heart Rate	Daily Steps	Sleep Disorder
16	80	4000	Sleep Apnea
18	80	4000	Insomnia
30	78	4100	Sleep Apnea
31	78	4100	Insomnia
32	69	6800	NaN
..	...	...	...
369	68	7000	Sleep Apnea
370	68	7000	Sleep Apnea
371	68	7000	Sleep Apnea
372	68	7000	Sleep Apnea
373	68	7000	Sleep Apnea

[185 rows x 13 columns]

```
sorted_df = df.sort_values(
    by="Sleep Duration",
    ascending=False
)
print(sorted_df)
```

	Person ID	Gender	Age	Occupation	Sleep Duration \
329	330	Female	53	Engineer	8.5
330	331	Female	53	Engineer	8.5
325	326	Female	53	Engineer	8.5
327	328	Female	53	Engineer	8.5
320	321	Female	53	Engineer	8.5
..	...	...	...	...	...
5	6	Male	28	Software Engineer	5.9
4	5	Male	28	Sales Representative	5.9
3	4	Male	28	Sales Representative	5.9
81	82	Female	34	Scientist	5.8
80	81	Female	34	Scientist	5.8

	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category \
329	9	30	3	Normal
330	9	30	3	Normal
325	9	30	3	Normal
327	9	30	3	Normal
320	9	30	3	Normal
..	...	...	...	...
5	4	30	8	Obese
4	4	30	8	Obese
3	4	30	8	Obese
81	4	32	8	Overweight
80	4	32	8	Overweight

	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
329	125/80	65	5000	NaN
330	125/80	65	5000	NaN
325	125/80	65	5000	NaN
327	125/80	65	5000	NaN
320	125/80	65	5000	NaN
..	...	...	...	...
5	140/90	85	3000	Insomnia
4	140/90	85	3000	Sleep Apnea
3	140/90	85	3000	Sleep Apnea
81	131/86	81	5200	Sleep Apnea
80	131/86	81	5200	Sleep Apnea

[374 rows x 13 columns]

```
print("Mean =",
      df["Sleep Duration"].mean())
print("Median =",
      df["Sleep Duration"].median())
print("Mode =")
print(df["Sleep Duration"].mode())
print("Standard Deviation =",
      df["Sleep Duration"].std())
```

```

Mean = 7.132085561497325
Median = 7.2
Mode =
0    7.2
Name: Sleep Duration, dtype: float64
Standard Deviation = 0.7956567308898189

```

```
print(df.describe())
```

	Person ID	Age	Sleep Duration	Quality of Sleep \
count	374.000000	374.000000	374.000000	374.000000
mean	187.500000	42.184492	7.132086	7.312834
std	108.108742	8.673133	0.795657	1.196956
min	1.000000	27.000000	5.800000	4.000000
25%	94.250000	35.250000	6.400000	6.000000
50%	187.500000	43.000000	7.200000	7.000000
75%	280.750000	50.000000	7.800000	8.000000
max	374.000000	59.000000	8.500000	9.000000

	Physical Activity Level	Stress Level	Heart Rate	Daily Steps
count	374.000000	374.000000	374.000000	374.000000
mean	59.171123	5.385027	70.165775	6816.844920
std	20.830804	1.774526	4.135676	1617.915679
min	30.000000	3.000000	65.000000	3000.000000
25%	45.000000	4.000000	68.000000	5600.000000
50%	60.000000	5.000000	70.000000	7000.000000
75%	75.000000	7.000000	72.000000	8000.000000
max	90.000000	8.000000	86.000000	10000.000000

```

gender_avg = df.groupby(
    "Gender"
)["Sleep Duration"].mean()
print(gender_avg)

```

```

Gender
Female    7.229730
Male      7.036508
Name: Sleep Duration, dtype: float64

```

```

corr = df[[
    "Sleep Duration",
    "Quality of Sleep"
]].corr()
print(corr)

```

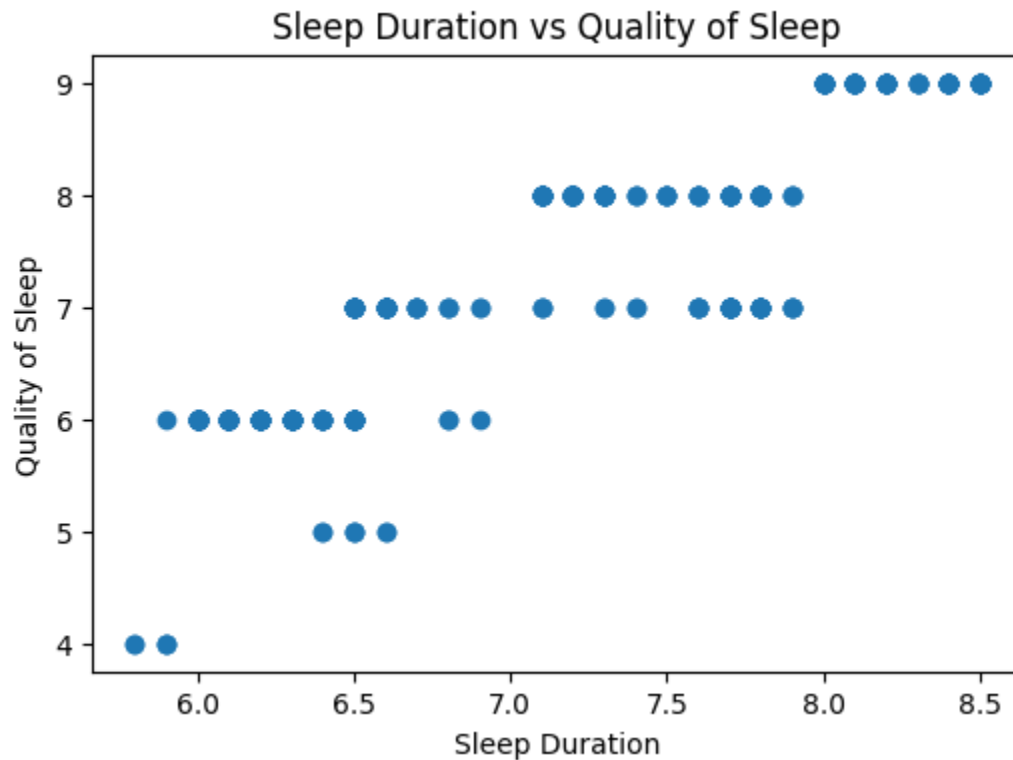
	Sleep Duration	Quality of Sleep
Sleep Duration	1.000000	0.883213
Quality of Sleep	0.883213	1.000000

```

import matplotlib.pyplot as plt
plt.figure(figsize=(6,4))

```

```
plt.scatter(
    df["Sleep Duration"],
    df["Quality of Sleep"]
)
plt.xlabel("Sleep Duration")
plt.ylabel("Quality of Sleep")
plt.title("Sleep Duration vs Quality of Sleep")
plt.show()
```



```
import matplotlib.pyplot as plt
top = df.sort_values(
    by="Sleep Duration",
    ascending=False
).head(10)
plt.figure(figsize=(8,5))
plt.bar(top["Person ID"].astype(str),
        top["Sleep Duration"])
plt.xlabel("Person ID")
plt.ylabel("Sleep Duration")
plt.title("Top 10 Sleep Duration")
plt.show()
```

